



OPEN High variability in carbon stable isotopes in the macroalga *Ulva* spp. and implications for eutrophication indicators

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Ulva spp. are among the most commonly encountered macroalgal species forming algal blooms and are responsible for many Green Tide events worldwide. Numerous adaptations have provided this genus with broad plasticity in carbon acquisition and metabolism, allowing rapid responses to nutrient inputs and sustained growth during the limiting conditions that arise during large bloom events. Because of this plasticity, the carbon isotopic composition of *Ulva* spp. may be highly variable. A compilation of 1,382 observations of *Ulva* $\delta^{13}\text{C}$ signatures from literature sources and systematic surveys revealed an unusually wide range of values, much broader than what is typically reported for other marine macroalgae, with a bimodal distribution. This pattern suggests two distinct growth modes in *Ulva* spp.: one characteristic of non-limiting conditions, and another associated with bloom conditions where rapid growth and carbon limitation drive higher $\delta^{13}\text{C}$ signatures. Analysis of environmental drivers in New England sites further showed that enriched *Ulva* $\delta^{13}\text{C}$ values arise under the limiting conditions that develop within dense macroalgal canopies in eutrophic estuaries. Excess nutrient inputs stimulate large macroalgal accumulations, which then alter the surrounding physicochemical environment and shift isotopic signatures. These results confirm that *Ulva* $\delta^{13}\text{C}$ is a valuable tool for assessing the trophic status of coastal environments.

Keywords *Ulva*, Carbon stable isotopes, Eutrophication, Algal blooms, Indicators

In the coastal ocean, primary production is often limited by the availability of nutrients. The growing human population and increased settlement and use of coastal areas have caused a worldwide rise in nutrient inputs to coastal waters^{1,2}. This increased nutrient availability often leads to the overgrowth of primary producers, which in turn can cause decreased water quality and eutrophication, a growing threat to many coastal ecosystems³.

In many shallow coastal areas, a common symptom of eutrophication is the occurrence of marine macroalgal blooms^{4–8}. The effects of these blooms on receiving environments include drastic restructuring of natural communities and disruption of ecosystem function^{7,9–11}. Accumulations of seaweed can physically damage other coastal life in valuable ecosystems such as seagrass beds^{7,12,13}, coral reefs^{14,15}, or salt marshes¹⁶. The decay of excessive algal organic matter can also foster anoxic conditions that lead to fish and shellfish kills^{17–19}, threatening commercial harvests. Other consequences of macroalgal blooms make coastal areas increasingly undesirable for human use, for example, by decreasing aesthetic value or affecting navigation^{20,21}, impacts that can also result in economic losses²². Additionally, large amounts of decomposing algal material can accumulate along shorelines, producing bad smells or even more severe human health issues²³.

Macroalgal blooms typically involve relatively few opportunistic taxa that are widely distributed along the world's coasts^{7,24}. Although various taxonomic groups of green, brown and red macroalgae can form blooms, in many cases the problem is characterized by large accumulations of green algae along the beaches of shallow coastal ecosystems, often referred to as “Green Tides”.

Among the proliferating green algae, species of the genus *Ulva* are the most commonly encountered^{24,25}. *Ulva* spp. are responsible for many Green Tide events worldwide, including the massive free-floating blooms of *Ulva prolifera* in the Yellow Sea, the largest Green Tides ever recorded²⁶. Due to their particular morphological and physiological adaptations, *Ulva* spp. respond rapidly to nutrient inputs with high rates of nutrient uptake and growth^{27,28}. The key to *Ulva*'s success lies in its ability to rapidly accumulate biomass under suitable environmental conditions, often outcompeting other fast-growing marine plants and algae^{29–31}. As bloom-forming species,

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Ulva spp. frequently face strong inorganic carbon limitation due to rapid growth and high population densities. Additionally, when *Ulva* spp. grow in dense blooms, the formation of stagnant boundary layers around algal tissues can restrict inorganic carbon diffusion. The species of this genus must have evolved a variety of carbon acquisition strategies to overcome these limitations³². These include the ability to access bicarbonate from seawater as an inorganic carbon source through carbon concentrating mechanisms that convert it to CO₂ for photosynthesis^{32–34}, as well as plasticity in use of different photosynthetic carbon metabolic pathways^{35–37}.

A common approach used to identify carbon sources and metabolic pathways in plants and algae is by analyzing the ratio of the two stable isotopes of carbon. The stable carbon isotope composition of plants and algae is a useful tool for understanding physiological processes and their responses to varying environmental conditions. Carbon has two naturally occurring stable isotopes, ¹²C and ¹³C. 98.9% of naturally occurring carbon is ¹²C, while ¹³C is much less abundant, accounting for only about 1.1%³⁸. Because biological and chemical processes typically prefer ¹²C but retain ¹³C, ecologists can examine the ratio of heavy to light isotopes to gain insights into carbon sources and metabolic processes within plant and algal tissues. The ratios are expressed in delta notation (δ), where $\delta^{13}\text{C} (\text{‰}) = [(R_{\text{sample}} - R_{\text{standard}})/R_{\text{standard}}] \cdot 1000$, with R representing the ratio of heavy to light isotopes in a sample relative to a standard. Previous compilations of $\delta^{13}\text{C}$ values and ranges in marine macroalgae have shown that most genera exhibit very distinct isotopic values, typically constrained within narrow ranges of less than 10‰ within a species³⁹.

Given the plasticity in carbon acquisition and metabolisms discussed above, it might be expected that the isotopic composition of *Ulva* spp. tissues spans a wide range of $\delta^{13}\text{C}$ values. By examining the variability in carbon isotopic signatures of *Ulva* spp., one of the most prominent and widespread algae causing Green Tides, we can gain insight into the traits that make this genus so successful at exploiting excess nutrients and forming large, often disruptive, blooms. In this paper, available literature data on stable carbon isotope ratios in *Ulva* spp. were reviewed, with particular emphasis on the factors relevant to observed variation in isotope signatures. Next, a comprehensive dataset from macroalgal surveys conducted at a subset of New England sites was analyzed to elucidate the relative influence of drivers responsible for variability in $\delta^{13}\text{C}$ signatures in *Ulva* spp. Finally, the potential use of stable carbon isotopes in *Ulva* spp. for studies of eutrophication and algal blooms was addressed.

Methods

Literature data collection

An extensive literature search using online tools and bibliographic databases, including Web of Science (<http://isiknowledge.com>), and Google Scholar (<http://scholar.google.com>), was conducted by searching the keywords “stable isotope” AND “*Ulva* OR Enteromorpha”. Results from this search were carefully screened to ensure included data was relevant to this study. Several criteria were used to determine whether data could be included. Only full-tissue, naturally occurring $\delta^{13}\text{C}$ algal signatures were used. Data on compound-specific stable isotope analyses, experimental manipulations and isotope enrichment experiments were excluded. When available, unique individual observations were used, but if data were aggregated and reported as averages, they were included only when the number of samples was small and collected over a relatively limited spatial area. A total of 188 publications, including 173 scientific journal articles, 10 theses, and 5 technical reports published between 1964 and 2025, were selected. The complete list of literature sources is included as Supplementary Materials. Data presented in graphs were extracted by using the WebPlotDigitizer V5 software. Information on the sampling site location and typology was also collected.

Linking environmental variables to *Ulva* $\delta^{13}\text{C}$ variability

To elucidate the relative influence of various environmental drivers responsible for variability in $\delta^{13}\text{C}$ signatures in *Ulva* spp., a comprehensive dataset from macroalgal and water quality surveys conducted in a subset of New England sites was analyzed. A collection of sites within the Waquoit Bay estuarine system on Cape Cod, Massachusetts, were selected for this study. The shallow system, with most areas less than 1 m deep, regularly experiences the development of dense macroalgal canopies⁷. Various species of bloom forming algae, including *Ulva lactuca*, are common in the area⁴⁰.

Interestingly, the various small sub-estuaries entering Waquoit Bay serve as a landscape-scale experiment on the effects of estuarine eutrophication. The surrounding sub-watersheds exhibit contrasting land cover and land use, ranging from protected parkland to densely developed suburban areas. These differences result in variable nitrogen discharges to the sub-estuaries^{40,41}, with annual loads ranging from 5 to 580 kg N per hectare of estuary area, levels comparable to those experienced by over 90% of the world's estuaries⁴². The wide range of nitrogen load magnitudes entering the various Waquoit Bay sub-estuaries help explain the dissimilarities in water quality and macroalgal biomass at different locations^{7,40,42}.

Macroalgal samples were collected intermittently from sites within the Waquoit Bay estuarine system between 2015 and 2024. Samples were collected using a 152 × 152 mm Ekman grab, rinsed through a 1 mm mesh to remove excess mud and sediment, and stored on ice until further processing in the lab. In the laboratory, samples were sorted by species. All algal matter was then dried at 60 °C to constant weight for at least 24 h before being weighed. Algal biomass was expressed as grams of dry tissue per square meter of estuary bottom. Dried macroalgal samples were ground and analyzed for carbon stable isotope signatures. The stable isotopic values were determined using a Europa 20–20 Continuous-Flow Isotope Ratio Mass Spectrometer (Sercon Ltd., Cheshire, UK).

Water samples from the same sites were collected by submerging an acid-washed 1-L polypropylene bottle below the water surface. 60 mL of the samples was immediately filtered through a 25-mm diameter, pre-ashed GF/F Whatman microfiber filter and frozen for storage and transport to the laboratory. In the laboratory, nitrate + nitrite concentrations were determined on a Lachat QuikChem 8000 AutoAnalyzer (PerkinElmer, Shelton, Connecticut) using a standard colorimetric assay⁴³. Ammonium concentrations were determined using the

Solórzano method⁴⁴ and measured using a Varian Cary-50 UV-Visible Spectrophotometer (Varian Inc., Palo Alto, California). In this study, the concentrations (μM) of nitrate, nitrite and ammonium in estuarine waters were summed and reported as Dissolved Inorganic Nitrogen (DIN).

Data on water temperature ($^{\circ}\text{C}$) and pH, variables selected to be potential drivers responsible for variability in *Ulva* $\delta^{13}\text{C}$ signatures, were obtained from the Waquoit Bay National Estuarine Research Reserve System-wide Monitoring Program. The Program collects continuous 15-min basic water quality data logging. Protocols are standardized and require frequent calibration and redeployment of YSI sondes⁴⁵. For the purpose of this study, all available temperature and pH observations on the 15 days before each of the macroalgal sampling dates were averaged, assuming the conditions during those 15 days were representative of the environment in which algal tissues developed.

Additionally, recently published data on annual nitrogen loads (expressed as kg N ha^{-1}) entering the various Waquoit Bay sub-estuaries were also compiled⁴⁰. Since DIN data only provide a snapshot of nutrient concentrations at the time of sampling, annual nitrogen loads offer a more accurate representation of total nutrient availability for algal growth throughout the entire growing season.

Structural equation modeling (SEM) was used to explore the direct and indirect effects of water temperature, pH, nutrient availability, and algal biomass on *Ulva* $\delta^{13}\text{C}$ signatures. The model was built based on hypothesized causal pathways that link nitrogen inputs from watersheds to changes in water chemistry and algal responses.

The SEM was fitted using the *lavaan* package⁴⁶ in R. The model included several predictor variables: nitrogen loading, algal biomass, pH, and DIN. Log transformations of nitrogen loading and DIN data, and square-root of algal biomass were applied to improve linearity and meet model assumptions. Univariate normality was tested with Shapiro-Wilk tests. The model was fitted to 62 complete observations. Standardized path coefficients were used to assess the relative strength of relationships among variables, and the model fit was evaluated using several indices such as the Chi-square (χ^2) test, Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR). Residual normality of $\delta^{13}\text{C}$ was assessed using a Shapiro-Wilk test.

Results and discussion

Global variability in *Ulva* $\delta^{13}\text{C}$ signatures

The compilation of *Ulva* $\delta^{13}\text{C}$ signatures done in this work contained a total of 1,382 observations. The dataset includes observations of at least 16 different *Ulva* species (Table 1) collected in 49 countries on the coasts of all continents and major regional seas (Fig. 1).

Compiled *Ulva* $\delta^{13}\text{C}$ signatures revealed a remarkable wide range of values, from a minimum of -26.7‰ to a maximum of -3.9‰ (Fig. 2). The observed range is much broader than what is typically reported for most marine macroalgae, which usually show ranges of less than 10‰ ³⁹. The $\delta^{13}\text{C}$ values displayed a bimodal distribution, with a relatively more frequent group of observations centered around the most ^{13}C -depleted isotopic signatures (mean $\approx -16.2\text{‰}$), and a less frequent but still abundant group of ^{13}C -enriched signatures (mean $\approx -10.9\text{‰}$; Fig. 2).

Compiled $\delta^{13}\text{C}$ values did not display any apparent relationship with latitude (Fig. 1 inset), and neither the width of the $\delta^{13}\text{C}$ range nor the existence of the two modes appeared to be associated with species-specific differences in signatures (Table 1). This suggests that the most likely explanation for the bimodal pattern is the existence of two distinct modes of carbon acquisition and/or metabolism in the genus.

Type of thallus	Species name (synonyms)	# of observations	$\delta^{13}\text{C}$ range
Blade	<i>Ulva australis</i> (<i>U. pertusa</i>)	42	-24.0 to -4.7
	<i>Ulva batuffolosa</i>	13	-21.6 to -10.0
	<i>Ulva conglobata</i>	3	-18.7 to -16.3
	<i>Ulva expansa</i>	6	-19.9 to -17.4
	<i>Ulva lactuca</i> (<i>U. fasciata</i> , <i>U. rutundea</i>)	339	-26.6 to -4.9
	<i>Ulva ohnoi</i>	6	-16.2 to -9.7
	<i>Ulva reticulata</i>	5	-26.7 to -17.5
	<i>Ulva rigida</i>	41	-21.8 to -8.8
	<i>Ulva taeniata</i>	1	-17.8
	appear as <i>Ulva</i> sp.	732	-25.9 to -3.9
Tubular	<i>Ulva clathrata</i> (<i>U. crinita</i>)	6	-14.6 to -12.0
	<i>Ulva compressa</i>	4	-17.8 to -15.0
	<i>Ulva flexuosa</i>	1	-20.1
	<i>Ulva intestinalis</i>	77	-23.9 to -8.3
	<i>Ulva linza</i>	5	-16.8 to -12.5
	<i>Ulva marginata</i>	1	-16.6
	<i>Ulva prolifera</i> (<i>U. salina</i>)	26	-23.2 to -11.8
	appear as <i>Enteromorpha</i> sp.	76	-20.9 to -5.5

Table 1. $\delta^{13}\text{C}$ ranges for the various *Ulva* species found in the literature and macroalgal surveys.

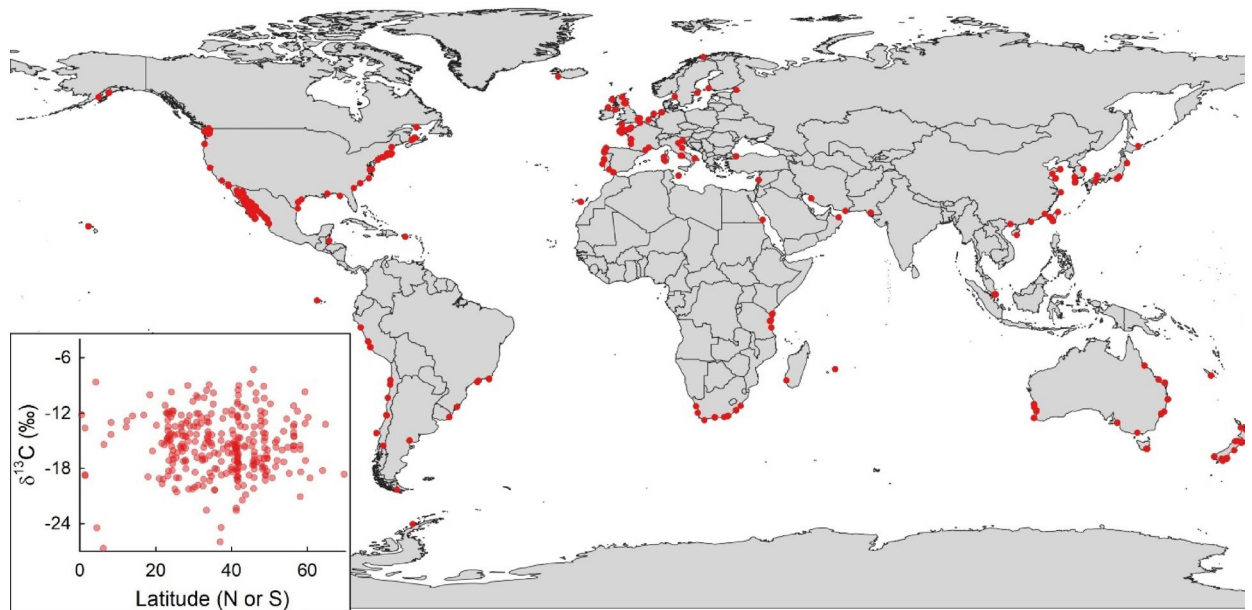


Fig. 1. Map of sites where data on *Ulva* $\delta^{13}\text{C}$ signatures were compiled from the literature. The inset shows the relationship between reported $\delta^{13}\text{C}$ values and latitude. The map was generated using QGIS v3.40.7.

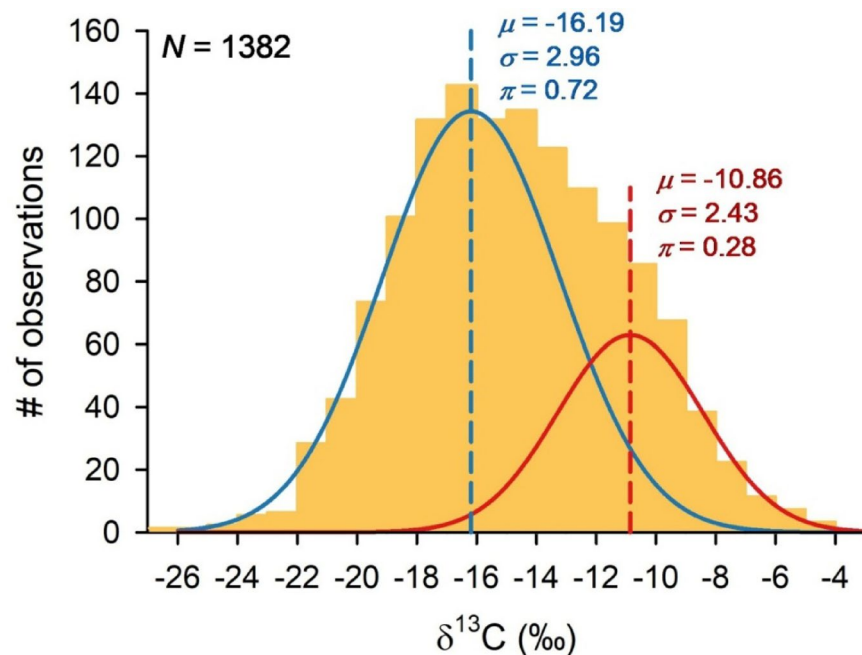


Fig. 2. A histogram representing the distribution of $\delta^{13}\text{C}$ values in *Ulva* spp. samples collected around the world. The curves in the graph represent the results of a mixture model analysis, including values for the mean (μ), standard deviation (σ), and probability (π) of the two distributions.

Several factors could strongly influence the isotopic make up of *Ulva* spp. tissues. First, the formation of diffusive boundary layers around algal tissues, especially in areas with restricted water exchange or dense algal accumulations, can limit inorganic carbon diffusion. This can lead to the entrapment of ^{13}C , which is otherwise more strongly discriminated against, resulting in elevated $\delta^{13}\text{C}$ values in algal tissue⁴⁷. Second, the ability of *Ulva* spp. to use bicarbonate instead of CO_2 as an inorganic carbon source^{34,39}, may also lead to higher $\delta^{13}\text{C}$ values, both because bicarbonate has a more enriched isotopic signature than dissolved CO_2 , and because the processes associated with active transport of bicarbonate across algal cell membranes produce smaller isotopic fractionation than the diffusive transport of CO_2 ⁴⁸. Third, the presence of both biophysical and biochemical

carbon concentration mechanisms in *Ulva* spp., and metabolic plasticity to use different photosynthetic carbon pathways, including coexisting C_3 and C_4 metabolism^{32,35–37,49}, can also contribute to increased $\delta^{13}C$ values, as enhanced carbon acquisition rates reduce ^{13}C discrimination.

Limited diffusion of inorganic carbon, the ability to use bicarbonate as a carbon source, and physiological adaptations that maintain high carbon uptake rates are all conditions commonly experienced by *Ulva* spp. during bloom events. It is therefore reasonable to infer that the observed bimodal distribution of $\delta^{13}C$ values reflects two different modes of growth: one more typical under non-limiting conditions, and another associated with bloom conditions, where rapid growth and carbon limitation drive higher $\delta^{13}C$ signatures in the algal tissues.

When information on the type of coastal environment where *Ulva* spp. specimens were collected was considered, an emergent pattern became evident. Although the ranges in *Ulva* $\delta^{13}C$ values were similar across all coastal environments, higher values (i.e. those centered around -10.9‰) were more frequently observed in environments with more restricted water exchange such as shallow coastal lagoons, semi-enclosed bays and harbors, and estuaries. In contrast, samples from more exposed environments, including wave-exposed beaches and island shorelines, more consistently displayed lower $\delta^{13}C$ values, with fewer occurrences of elevated signatures (Fig. 3). Coastal systems with intermediate levels of water exchange, such as rias, fjords, and open bays or gulfs, exhibited an intermediate range of $\delta^{13}C$ values (Fig. 3). This pattern suggests that the environmental conditions conducive to elevated $\delta^{13}C$ signatures in *Ulva* spp., such as carbon limitation and altered inorganic carbon availability, are more prevalent in settings where water exchange is restricted.

Interestingly, the variability in $\delta^{13}C$ signatures of *Ulva* spp. differed significantly across thallus morphologies (blades vs. tubular forms; Mann–Whitney $U=149,267$, $p<0.001$). Although the range of $\delta^{13}C$ values was similarly broad for both morphologies (Table 1), enriched $\delta^{13}C$ signatures were more frequently observed in blade-like species. Despite the statistical significance of this difference, the effect size was small (rank-biserial $r=0.18$; Cliff's $\delta=0.30$), indicating that the magnitude of this variation across morphologies was modest. Several factors may have limited the conclusiveness of these results. For instance, there was a substantial disparity in data availability between the two growth forms, with six times more data from blade-like species than from

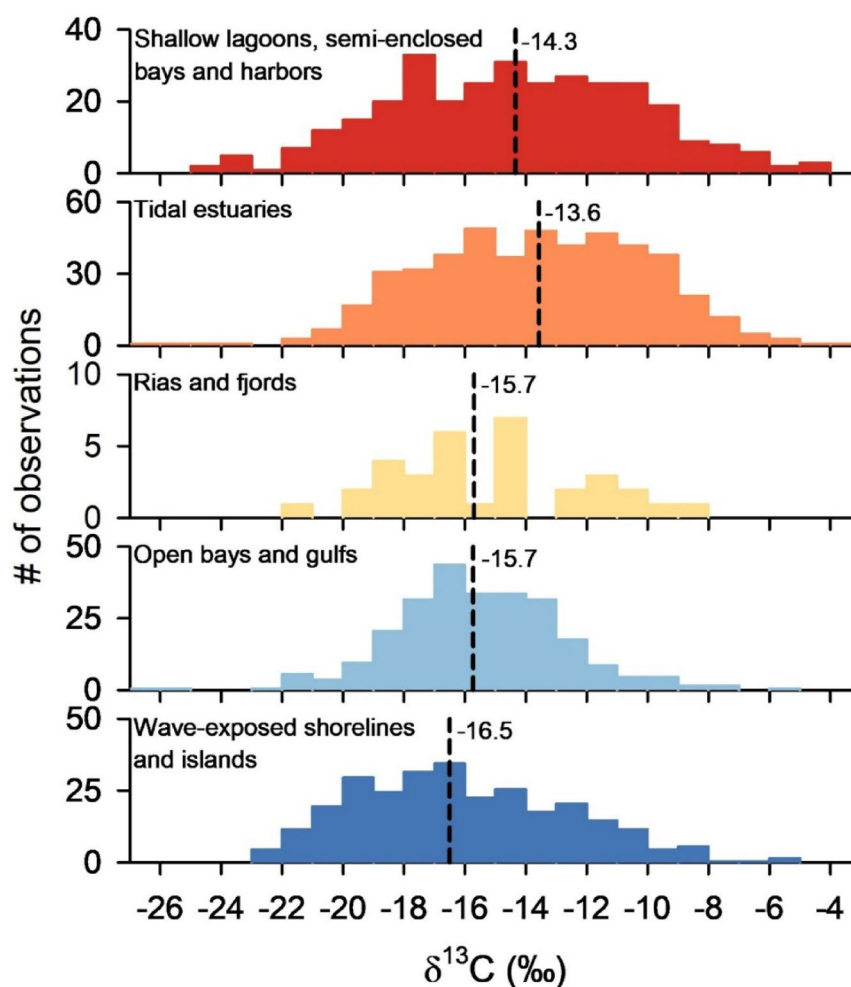


Fig. 3. Histograms representing the distribution of $\delta^{13}C$ values in *Ulva* spp. samples collected in various types of coastal environments around the world. The dashed lines and numbers in the graph represent the median on each environment.

tubular ones. Another influencing factor could be the higher relative number of blade-like species observed in estuaries, environments where elevated $\delta^{13}\text{C}$ signatures are more common, compared to tubular species. Nonetheless, these findings suggest that, since blades and tubular forms differ in their surface area-to-volume ratios, morphology plays a role in the acquisition of inorganic carbon by algae during bloom conditions, thereby potentially shaping their isotopic composition.

Environmental drivers of *Ulva* $\delta^{13}\text{C}$ variability

The sub-estuaries of Waquoit Bay harbor a complex and abundant macroalgal community. The macroalgal canopy is composed of various opportunistic seaweed species growing together on the estuary bottoms, although occasionally large mats of floating macroalgae can also be observed in some locations. The majority of the biomass in the collected samples was contributed by the red algae *Gracilaria tikvahiae*, *Polysiphonia urceolata* and *Ceramium* sp., as well as the green algae *Cladophora vagabunda*, *Chaetomorpha* sp. and *Ulva lactuca*.

For most species, $\delta^{13}\text{C}$ analysis of algal tissues revealed distinct isotopic signatures with relatively narrow ranges (Fig. 4). In contrast, *U. lactuca* displayed a much wider range of $\delta^{13}\text{C}$ values (Fig. 4), despite co-occurring with other species within the same canopy and being exposed to similar environmental conditions. This suggests that *U. lactuca* is more responsive to environmental variation and its physiology leads to greater isotopic variability than the other algal groups.

To explore the drivers of this variability, the relationships between *Ulva* $\delta^{13}\text{C}$ values and a set of environmental variables measured across Waquoit Bay were examined. A positive correlation with temperature was initially expected, under the assumption that higher temperatures would promote faster growth, potentially leading to carbon limitation and enrichment in heavier carbon isotopes. However, the data did not support this hypothesis, and no clear relationship between temperature and $\delta^{13}\text{C}$ values was observed (Fig. 5). Similarly, an increase in *Ulva* $\delta^{13}\text{C}$ values at higher pH was anticipated, which could reflect decreased CO_2 availability and a greater reliance on bicarbonate as a carbon source. Yet again, no consistent trend emerged from the data (Fig. 5).

In contrast, nutrient availability appeared to have a clearer influence on *Ulva* $\delta^{13}\text{C}$. While relationships between $\delta^{13}\text{C}$ and water column DIN at the time of sampling were weak, stronger patterns emerged when $\delta^{13}\text{C}$

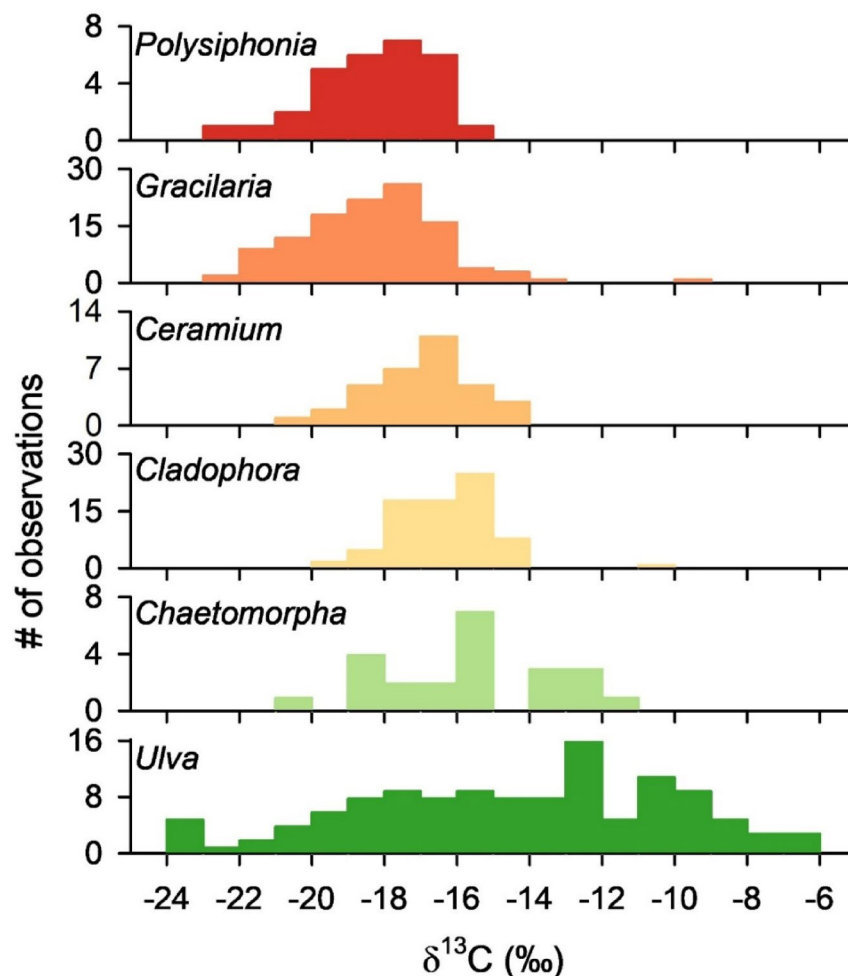


Fig. 4. Histograms representing the distribution of $\delta^{13}\text{C}$ values in the most abundant genus of algae found in Waquoit Bay sub-estuaries.

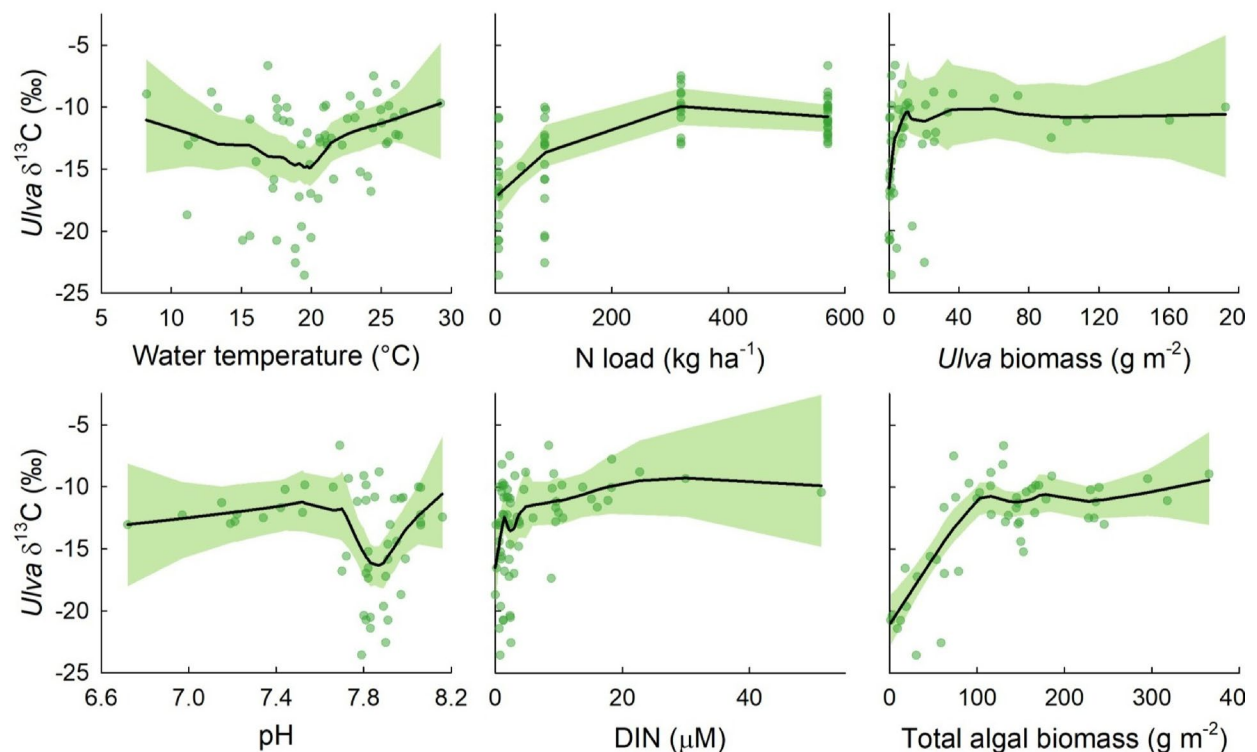


Fig. 5. Relationships between *Ulva* $\delta^{13}\text{C}$ and various environmental variables across study sites in Waquoit Bay sub-estuaries. Each panel shows a scatterplot of individual observations, a locally weighted regression (LOESS, solid line), and 95% confidence intervals (shaded areas) calculated by bootstrapping.

values were compared to total nitrogen load to each sub-estuary (Fig. 5). These results suggest that chronic nitrogen enrichment, rather than short-term DIN availability, may drive faster growth rates in *Ulva* spp., leading to isotopic enrichment.

The strongest and most consistent patterns in *Ulva* $\delta^{13}\text{C}$ emerged when compared to algal biomass. Surprisingly, these patterns were more robust when total macroalgal biomass, including all canopy species, was considered, rather than *U. lactuca* biomass alone. At sites with low *U. lactuca* biomass, $\delta^{13}\text{C}$ values ranged widely from depleted to enriched. However, at sites with high *U. lactuca* biomass, $\delta^{13}\text{C}$ values were consistently higher (Fig. 5). This pattern was even more pronounced when total algal biomass (all species combined) was used: low biomass corresponded with depleted $\delta^{13}\text{C}$, while high biomass was associated with enriched and stable $\delta^{13}\text{C}$ signatures (Fig. 5).

These results suggest that the broader macroalgal community, not just *Ulva* spp., plays an important role in shaping the biogeochemical conditions that influence *Ulva* $\delta^{13}\text{C}$. High overall algal biomass reflects eutrophic conditions that promote rapid growth and carbon limitation, ultimately driving the observed isotopic shifts.

To further explore the influence of environmental drivers on *Ulva* $\delta^{13}\text{C}$, and to assess the relative importance of different mechanisms underlying its variability, a SEM, including four endogenous variables: pH, DIN, total algal biomass, and *Ulva* $\delta^{13}\text{C}$; and two exogenous variables: temperature and nitrogen load, was constructed (Fig. 6). Overall, the SEM showed good fit: $\chi^2(2) = 2.40$, $p = 0.302$; CFI = 0.997; TLI = 0.982; RMSEA = 0.056; SRMR = 0.038. The model explained 51% of the variance in $\delta^{13}\text{C}$ ($R^2 = 0.511$).

Path coefficients revealed several significant relationships consistent with the initial hypotheses. Nitrogen loading had strong positive effects on both algal biomass and DIN concentrations. In turn, algal biomass negatively affected pH, indicating that greater biomass reduces CO_2 availability in the system (Fig. 6). Algal biomass also had a positive effect on DIN, though this may reflect covariance between the two variables, as both increase with higher nitrogen inputs (Fig. 6). Water temperature had negative effects on both DIN and pH, suggesting that warmer periods lead to decreases in DIN concentrations due to increased nutrient demand by primary producers, as well as increases in ecosystem respiration and CO_2 production, which in turn drive pH values down (Fig. 6).

With respect to $\delta^{13}\text{C}$, algal biomass had a strong positive effect, supporting the hypothesis that higher biomass is associated with less depleted isotopic signatures (Fig. 6). This reinforces the idea that the development of dense macroalgal canopies influences biogeochemical conditions in ways that very markedly shape *Ulva* $\delta^{13}\text{C}$.

Ulva $\delta^{13}\text{C}$ also appeared to be negatively affected by pH (Fig. 6). Elevated pH is often associated with heavier $\delta^{13}\text{C}$ signatures in marine primary producers due to reduced availability of dissolved CO_2 and greater reliance on bicarbonate⁴⁸. Nevertheless, several factors could explain the absence of a clear relationship in this case. First, the pH measurements used here represent conditions in the upper part of the water column, which may not accurately reflect the microenvironment experienced by *U. lactuca* in the macroalgal mats near the estuary

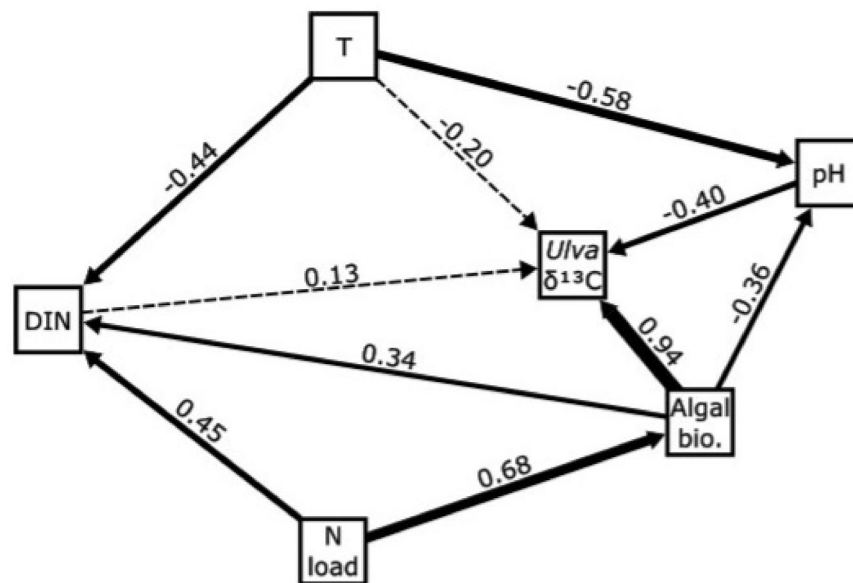


Fig. 6. Structural equation model (SEM) showing hypothesized relationships among nitrogen loading (N load, Log transformed), algal biomass (Sqrt transformed), dissolved inorganic nitrogen (DIN, Log transformed), pH, temperature (T), and carbon isotope composition of *Ulva lactuca* (*Ulva* $\delta^{13}\text{C}$). Arrows represent significant pathways (solid lines) and non-significant pathways (dashed lines). Standardized path coefficients are shown along each arrow, with thickness proportional to effect size.

bottom. Dense macroalgal canopies can create steep vertical gradients in water chemistry, especially in these very shallow systems with limited vertical mixing, making surface pH a poor proxy for conditions within the canopy. Second, the use of averaged pH values likely masked short-term variability that can be ecologically significant. In systems with high algal biomass, pH often fluctuates dramatically over diel and tidal cycles, potentially exposing the algae to transient CO_2 -limiting conditions that could enrich $\delta^{13}\text{C}$. Additionally, recent experimental studies indicate that light, rather than pH or elevated CO_2 , is the primary driver of *Ulva* spp. photosynthesis and growth⁵⁰, suggesting that under dense macroalgal canopies, factors such as shading and local inorganic carbon availability, particularly the predominance of bicarbonate, are more likely to influence $\delta^{13}\text{C}$ variability than bulk water pH alone.

DIN and temperature did not have significant direct effects on *Ulva* $\delta^{13}\text{C}$ (Fig. 6). While both variables are known to influence algal growth rates, their variability alone was insufficient to explain the observed isotopic differences. This suggests that rapid growth, by itself, may not be enough to drive shifts in $\delta^{13}\text{C}$ signatures. Instead, it appears that the development of bloom conditions, characterized by high algal biomass and associated changes in water chemistry, is necessary to significantly alter *Ulva* spp.'s carbon isotopic composition.

Implications for eutrophication studies

Nitrogen stable isotopes are widely used to trace sources of excess nitrogen in eutrophication studies. Recent evidence suggests that $\delta^{13}\text{C}$ signatures of phytoplankton can also serve as sensitive indicators of eutrophication^{48,51}. The results presented here confirm that the same holds true for green macroalgae of the genus *Ulva*, a group of fast-growing opportunistic species that are among the most widespread contributors to Green Tide events globally.

The global data analysis highlights the broad applicability of using *Ulva* $\delta^{13}\text{C}$ as an indicator of eutrophication and macroalgal blooms. Elevated $\delta^{13}\text{C}$ values were more frequent in shallow, restricted coastal environments near human settlements—precisely the conditions where large nuisance macroalgal blooms are most frequent—underscoring this metric's potential for detecting eutrophic conditions.

Data from the Waquoit Bay sites further demonstrated that *Ulva* $\delta^{13}\text{C}$ is a responsive indicator of nutrient enrichment and macroalgal blooms. Together, the results support a mechanistic pathway in which land-derived nutrient inputs promote excessive macroalgal biomass, which in turn alters within-canopy water chemistry and shifts the carbon isotopic composition of *U. lactuca*. The lack of a clear direct response to changes in pH or temperature suggests that $\delta^{13}\text{C}$ signatures of *Ulva* spp. may not necessarily respond to conditions projected under future ocean acidification and warming scenarios. As observed in other algal species⁵², changes in seawater pH may not significantly affect the physiological processes of photosynthesis and nutrient uptake in *Ulva* spp. during bloom conditions. Instead, local conditions within dense algal canopies, shaped by factors such as nutrient enrichment and eutrophication, are likely to have a stronger influence on carbon isotope composition.

In many coastal systems, traditional indicators of eutrophication, such as nutrient concentrations or phytoplankton biomass, can be highly variable in space and time, often making the assessment of the trophic status of ecosystems difficult⁴². The results presented here confirm that *Ulva* $\delta^{13}\text{C}$ is a valuable tool for assessing

trophic status, as it responds sensitively to both nutrient enrichment and the development of excessive algal biomass, whether composed solely of *Ulva* spp. or mixed with other bloom-forming algae.

Data availability

Data from the extensive literature survey and the macroalgal surveys are provided in a supplementary data file included with the Supplementary Materials. Additional results from the analyses conducted during this study are available from the corresponding author upon reasonable request.

Received: 27 August 2025; Accepted: 15 October 2025

Published online: 19 November 2025

References

- Nixon, S. W. Coastal marine eutrophication: A definition, social causes, and future concerns. *Ophelia* **41**, 199–219 (1995).
- Valiela, I. *Global Coastal Change* (Blackwell Pub, 2006).
- Bricker, S. B. et al. Effects of nutrient enrichment in the nation's estuaries: A decade of change. *Harmful Algae* **8**, 21–32 (2008).
- Lavery, P. S., Lukateli, R. J. & McComb, A. J. Changes in the biomass and species composition of macroalgae in a eutrophic estuary. *Estuar. Coast Shelf Sci.* **33**, 1–22 (1991).
- Sfriso, A., Pavoni, B., Marcomini, A. & Orio, A. A. Macroalgae, nutrient cycles, and pollutants in the lagoon of Venice. *Estuaries* **15**, 517–528 (1992).
- Fox, S. E., Stieve, E., Valiela, I., Hauxwell, J. & McClelland, J. Macrophyte abundance in Waquoit Bay: Effects of land-derived nitrogen loads on seasonal and multi-year biomass patterns. *Estuaries Coasts* **31**, 532–541 (2008).
- Valiela, I. et al. Macroalgal blooms in shallow estuaries: Controls and ecophysiological and ecosystem consequences. *Limnol. Oceanogr.* **42**, 1105–1118 (1997).
- Morand, P. & Briand, X. Excessive growth of macroalgae: A symptom of environmental disturbance. *Bot. Mar.* **39**, 491–516 (1996).
- Lyons, D. A. et al. What are the effects of macroalgal blooms on the structure and functioning of marine ecosystems? A systematic review protocol. *Environ. Evid.* **1**, 1–6 (2012).
- Lapointe, B. E., Burkholder, J. A. M. & Van Alstyne, K. L. Harmful macroalgal blooms in a changing world: Causes, impacts, and management. In *Harmful Algal Blooms: A Compendium Desk Reference* (eds Shumway, S. E. et al.) 515–560 (Wiley, 2018). <https://doi.org/10.1002/9781118994672.CH15>.
- Lyons, D. A. et al. Macroalgal blooms alter community structure and primary productivity in marine ecosystems. *Glob Chang. Biol.* **20**, 2712–2724 (2014).
- McGlathery, K. J. Macroalgal blooms contribute to the decline of seagrass in nutrient-enriched coastal waters. *J. Phycol.* **37**, 453–456 (2001).
- Havens, K. E. et al. Complex interactions between autotrophs in shallow marine and freshwater ecosystems: implications for community responses to nutrient stress. *Environ. Pollut.* **113**, 95–107 (2001).
- McCook, L. J., Jompa, J. & Diaz-Pulido, G. Competition between corals and algae on coral reefs: A review of evidence and mechanisms. *Coral Reefs* **19**, 400–417 (2001).
- Hughes, T. P. Catastrophes, phase shifts, and large-scale degradation of a Caribbean coral reef. *Science* **265**, 1547–1551 (1994).
- Newton, C. & Thornber, C. Ecological impacts of macroalgal blooms on salt marsh communities. *Estuaries Coasts* **36**, 365–376 (2013).
- Burkholder, J. M. Beyond algal blooms, oxygen deficits and fish kills: Chronic, long-term impacts of nutrient pollution on aquatic ecosystems. In *Waters in Peril* (eds Bendell-Young, L. & Gallagher, P.) 103–125 (Springer, 2001). https://doi.org/10.1007/978-1-4615-1493-0_7.
- Diaz, R. J. Overview of hypoxia around the world. *J. Environ. Qual.* **30**, 275–281 (2001).
- Diaz, R. J. & Rosenberg, R. Spreading dead zones and consequences for marine ecosystems. *Science* **321**, 926–929 (2008).
- Fraga, J. & Robledo, D. Covid-19 and *Sargassum* blooms: Impacts and social issues in a mass tourism destination (Mexican Caribbean). *Marit Stud.* **21**, 159–171 (2022).
- Nilsson, J. H. & Gössling, S. Tourist responses to extreme environmental events: The case of Baltic Sea algal blooms. *Tour Plan. Dev.* **10**, 32–44 (2013).
- van Beukering, P. J. H. & Cesar, H. S. J. Ecological economic modeling of coral reefs: Evaluating tourist overuse at Hanauma Bay and algae blooms at the Kihiki Coast, Hawai'i. *Pac. Sci.* **58**, 243–260 (2004).
- Berdalet, E. et al. Marine harmful algal blooms, human health and wellbeing: Challenges and opportunities in the 21st century. *J. Mar. Biol. Assoc. U.K.* **96**, 61–91 (2016).
- Morand, P. & Merceron, M. Macroalgal population and sustainability. *J. Coast Res.* **21**, 1009–1020 (2005).
- Wang, H., Wang, G. & Gu, W. Macroalgal blooms caused by marine nutrient changes resulting from human activities. *J. Appl. Ecol.* **57**, 766–776 (2020).
- Liu, D. & Zhou, M. Green tides of the Yellow Sea: Massive free-floating blooms of *Ulva prolifera*. In *Global Ecology and Oceanography of Harmful Algal Blooms* (eds Glibert, P. M. et al.) 317–326 (Springer International Publishing, 2018). https://doi.org/10.1007/978-3-319-70069-4_16.
- Hernández, I., Peralta, G., Pérez-Lloréns, J. L., Vergara, J. J. & Niell, F. X. Biomass and dynamics of growth of *Ulva* species in Palmones River estuary. *J. Phycol.* **33**, 764–772 (1997).
- Geertz-Hansen, O., Sand-Jensen, K., Hansen, D. F. & Christiansen, A. Growth and grazing control of abundance of the marine macroalgae, *Ulva lactuca* L. in a eutrophic Danish estuary. *Aquat. Bot.* **46**, 101–109 (1993).
- Tang, Y. Z. & Gobler, C. J. The green macroalgae, *Ulva lactuca*, inhibits the growth of seven common harmful algal bloom species via allelopathy. *Harmful Algae* **10**, 480–488 (2011).
- Zeng, Y. et al. Review of allelopathy in green tides: The case of *Ulva prolifera* in the South Yellow Sea. *Biology* **13**, 456 (2024).
- Young, C. S., Peterson, B. J. & Gobler, C. J. The bloom-forming macroalgae, *Ulva*, outcompetes the seagrass, *Zostera marina*, under high CO₂ conditions. *Estuaries Coasts* **41**, 2340–2355 (2018).
- Sun, J. et al. Diversity of CO₂ concentrating mechanisms in macroalgae photosynthesis: A case study of *Ulva* sp. *J. Mar. Sci. Eng.* **11**, 1911 (2023).
- Drechsler, Z. & Beer, S. Utilization of inorganic carbon by *Ulva lactuca*. *Plant. Physiol.* **97**, 1439–1444 (1991).
- Larsson, C. & Axelsson, L. Bicarbonate uptake and utilization in marine macroalgae. *Eur. J. Phycol.* **34**, 79–86 (1999).
- Valiela, I., Liu, D., Lloret, J., Chenoweth, K. & Hanacek, D. Stable isotopic evidence of nitrogen sources and C4 metabolism driving the world's largest macroalgal green tides in the Yellow Sea. *Sci. Rep.* **8**, 17437 (2018).
- Zhang, X., Gao, G., Gao, Z., Gao, K. & Liu, D. The contribution of biophysical and biochemical CO₂ concentration mechanisms to the carbon fixation of the green macroalgae *Ulva prolifera*. *Mar. Life Sci. Technol.* **7**, 537–548. <https://doi.org/10.1007/s42995-024-00265-7> (2025).
- Liu, D. et al. Role of C4 carbon fixation in *Ulva prolifera*, the macroalgae responsible for the world's largest green tides. *Commun. Biol.* **3**, 1–8 (2020).

38. Fry, B. *Stable Isotope Ecology* (Springer New York, 2006). <https://doi.org/10.1007/0-387-33745-8>
39. Raven, J. A. et al. Mechanistic interpretation of carbon isotope discrimination by marine macroalgae and seagrasses. *Funct. Plant. Biol.* **29**, 355–378 (2002).
40. Valiela, I., Owens, C., Elmsstrom, E. & Lloret, J. Eutrophication of Cape Cod estuaries: Effect of decadal changes in global-driven atmospheric and local-scale wastewater nutrient loads. *Mar. Pollut. Bull.* **110**, 309–315 (2016).
41. Valiela, I. et al. Nitrogen loading from coastal watersheds to receiving estuaries: New method and application. *Ecol. Appl.* **7**, 358–380 (1997).
42. Valiela, I., Lloret, J., Chenoweth, K., Elmsstrom, E. & Hanacek, D. Control of N concentrations in Cape Cod estuaries by nitrogen loads, season, and down-estuary transit: Assessment by conventional and effect-size statistics. *Estuaries Coasts*. **44**, 1294–1309. <https://doi.org/10.1007/s12237-020-00869-z> (2021).
43. Wood, E. D., Armstrong, F. A. J. & Richards, F. A. Determination of nitrate in sea water by cadmium-copper reduction to nitrite. *J. Mar. Biol. Assoc. U.K.* **47**, 23–31 (1967).
44. Solorzano, L. Determination of ammonia in natural waters by the phenylhypochlorite method. *Limnol. Oceanogr.* **14**, 799–801 (1969).
45. Porter, D. E. et al. Data management in support of environmental monitoring, research, and coastal management. *J. Coast Res.* **45**, 9–16 (2004).
46. Rosseel, Y. Lavaan: an R package for structural equation modeling. *J. Stat. Softw.* **48**, 1–36 (2012).
47. France, R. L. Carbon-13 enrichment in benthic compared to planktonic algae: Foodweb implications. *Mar. Ecol. Prog. Ser.* **124**, 307–312 (1995).
48. Oczkowski, A., Markham, E., Hanson, A. & Wigand, C. Carbon stable isotopes as indicators of coastal eutrophication. *Ecol. Appl.* **24**, 457–466 (2014).
49. Xu, J. et al. Evidence of coexistence of C3 and C4 photosynthetic pathways in a green-tide-forming alga, *Ulva prolifera*. *PLoS One*. **7**, e37438 (2012).
50. Rautenberger, R. et al. Saturating light and not increased carbon dioxide under ocean acidification drives photosynthesis and growth in *Ulva rigida* (Chlorophyta). *Ecol. Evol.* **5**, 874–888 (2015).
51. Savage, C., Leavitt, P. R. & Elmgren, R. Effects of land use, urbanization, and climate variability on coastal eutrophication in the Baltic Sea. *Limnol. Oceanogr.* **55**, 1033–1046 (2010).
52. Fernández, P. A., Roldán, M. Y., Leal, P. P. & Hurd, C. L. Seawater pH, and not inorganic nitrogen source, affects pH at the blade surface of *Macrocystis pyrifera*: Implications for responses of the giant kelp to future oceanic conditions. *Physiol. Plant.* **159**, 107–119 (2017).

Acknowledgements

I am grateful to the personnel at the Waquoit Bay NERR for their assistance during the field work and for sharing their Waquoit Bay data records. I also want to thank the several cohorts of students from the Marine Biological Laboratory-Semester in Environmental Science and the many fellows from the Logan Science Journalism Program who contributed to the collection of samples in the Waquoit Bay sites.

Author contributions

JL: Conceptualization; Formal analysis; Investigation; Methodology; Visualization; Writing—original draft; Writing—review & editing.

Funding

This research did not receive any specific grant from funding agencies.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-025-24788-w>.

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